

# Customer Shopping Behavior Analysis

## 1. Project Overview

This project analyzes customer shopping behavior using Python, SQL, and Power BI. The dataset contains customer demographics, product categories, spending behavior, discount usage, shipping preferences, and review ratings. The goal is to identify business insights that can help improve customer engagement, optimize marketing strategies, and increase revenue.

## 2. Dataset Summary

| Metric                  | Value   |
|-------------------------|---------|
| Rows                    | 3900    |
| Columns                 | 18      |
| Average Purchase Amount | \$59.76 |
| Average Review Rating   | 3.75    |

## 3. Python Data Cleaning & Analysis

The dataset was cleaned and transformed using Python and Pandas. Main steps included: - Handling missing values - Checking duplicates - Feature engineering - Creating age groups - Exploratory Data Analysis - Database integration for PostgreSQL

### Python Code Snippet from Notebook

```
import pandas as pd

df = pd.read_csv(r'C:\Users\hp\Downloads\Customer Behavior Dashboard\customer_shopping_behavior.csv')

df.head()
```

## 4. SQL Business Analysis

SQL queries were written to answer important business questions related to: - Revenue analysis - Customer segmentation - Subscription behavior - Product performance - Discount analysis - Revenue contribution by age group

### SQL Query Snippets

```
create DATABASE customer_behavior;

SELECT * FROM customer;

# Q1. What is the total revenue generated by male vs. female customers?
SELECT
  gender,
  SUM(purchase_amount) as Total_Revenue
FROM customer
GROUP BY gender;

# Q2. Which customers used a discount but still spent more than the average purchase amount?
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = "Yes" and purchase_amount >= (SELECT AVG(purchase_amount) from customer);

# Q3. Which are the top 5 products with the highest average review rating?
SELECT item_purchased, Round(AVG(review_rating),2) as Avg_Rating
FROM customer
GROUP BY item_purchased
```

```
ORDER BY AVG(review_rating) DESC
limit 5;
```

```
# Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
SELECT shipping_type, round(AVG(purchase_amount),2) as Avg_purchase_amount
FROM customer
where shipping_type IN ("Express", "Standard")
GROUP BY shipping_type;
```

```
# Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.
SELECT subscription_status, COUNT(customer_id) as total_customers,
AVG(purchase_amount) as avg_spend, SUM(purchase_amount) as total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

```
# Q6. Which 5 products have the highest percentage of purchases with discounts applied?
SELECT item_purchased,
round( 100 * SUM(case when discount_applied = "Yes" then 1 else 0 end) / count(*), 2) as discount_rate
from customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

```
# Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and segment them into segments.
SELECT customer_segment,
COUNT(*) as customer_count
FROM (
  SELECT
  CASE
  WHEN previous_purchases = 1 THEN "New"
  WHEN previous_purchases BETWEEN 2 AND 10 THEN "Returning"
  ELSE "Loyal"
  END AS customer_segment
  FROM customer
) AS segments
GROUP BY customer_segment;
```

```
# Q8. What are the top 3 most purchased products within each category?
with item_counts as (
  SELECT category,
  item_purchased, COUNT(customer_id) as total_orders,
  ROW_NUMBER() over(PARTITION BY category order by count(customer_id) DESC) as item_rank
  from customer
  group by category, item_purchased
)

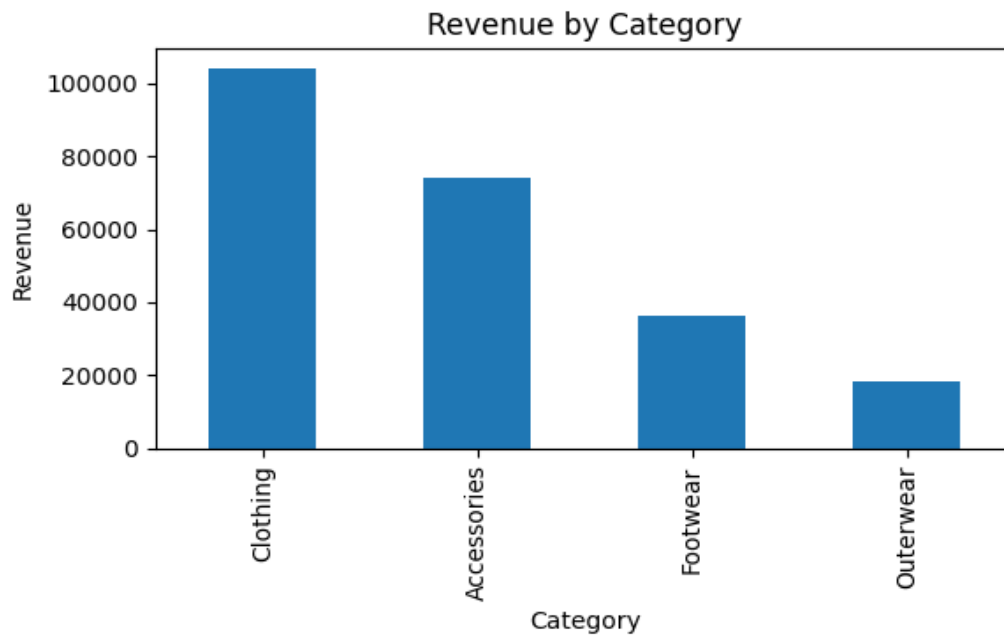
SELECT * FROM item_counts WHERE item_rank <= 3;
```

```
# Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
SELECT
subscription_status, COUNT(*) as repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

```
# Q10. What is the revenue contribution of each age group?
SELECT age_group,
SUM(purchase_amount) as Total_revenue
from customer
GROUP BY age_group
order by Total_revenue DESC;
```

## 5. Power BI Dashboard

The Power BI dashboard provides interactive insights into customer shopping behavior. It includes: - KPI Cards - Revenue by category - Subscription analysis - Sales by age group - Revenue by age group - Interactive filters



## 6. Business Recommendations

- Promote subscription benefits to improve retention - Reward repeat customers through loyalty programs - Focus marketing on high-revenue age groups - Highlight top-rated products in campaigns - Optimize discount strategy for profitability

## 7. GitHub Repository Structure

```
project-folder/  
├── data/  
│   └── customer_shopping_behavior.csv  
├── notebooks/  
│   └── Customer_Shopping_Behavior_Analysis.ipynb  
├── sql/  
│   └── customer_behavior.sql  
├── dashboard/  
│   ├── Customer Behavior Dashboard.pbix  
│   └── Customer Behavior Dashboard.pdf  
├── reports/  
│   ├── Business_Problem_Document_Final.pdf  
│   └── Customer_Shopping_Behavior_Project_Report.pdf  
└── README.md
```